

11. A planet of mass M and radius R rotates so rapidly that loose material at the equator just remains on the surface. What is the period of rotation of the planet?

G is the universal gravitational constant.

A $2\pi\sqrt{\frac{R}{GM}}$

B $2\pi\sqrt{\frac{R^2}{GM}}$

C $2\pi\sqrt{\frac{GM}{R^3}}$

D $2\pi\sqrt{\frac{R^3}{GM}}$

Answer: D

Gravitational force on material of mass m provides the centripetal force,

$$\frac{GMm}{R^2} = mR\omega^2$$

$$\frac{GM}{R^2} = R\left(\frac{2\pi}{T}\right)^2$$

$$T = 2\pi\sqrt{\frac{R^3}{GM}}$$